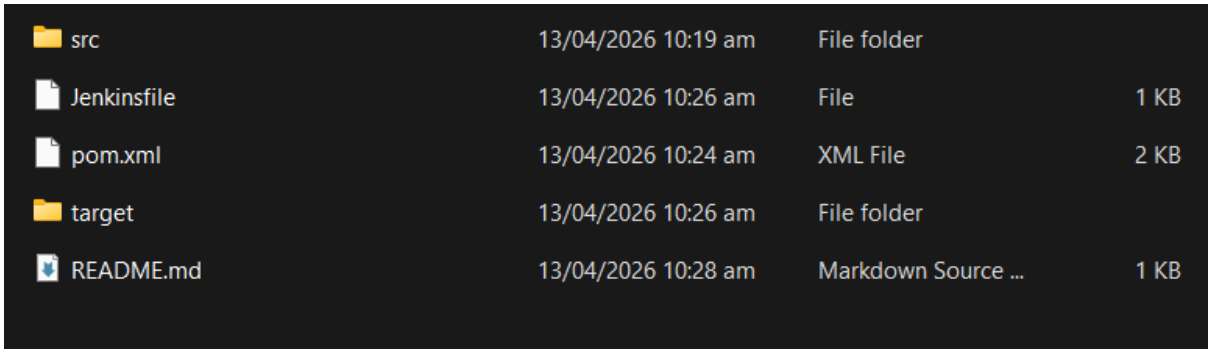
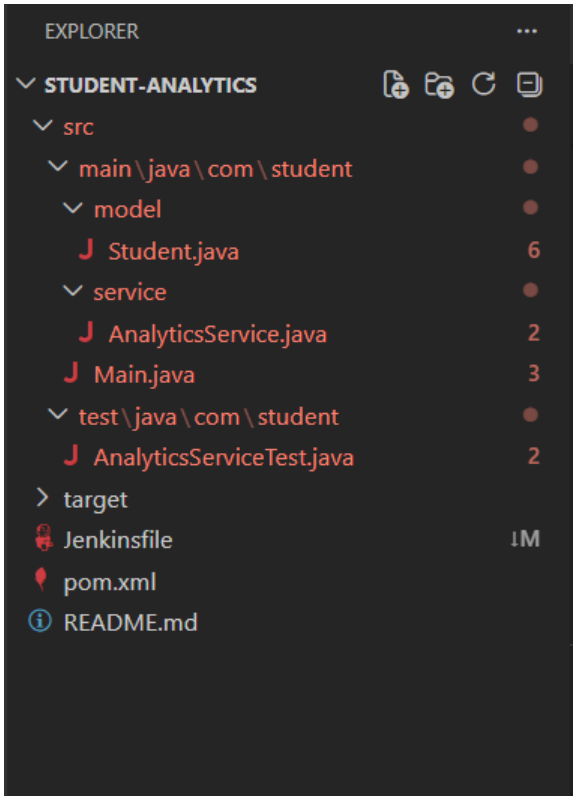
Agile Development and Process Management
LAB FAT
S Dhanush Kumar

Folder Structure:



src	13/04/2026 10:19 am	File folder	
Jenkinsfile	13/04/2026 10:26 am	File	1 KB
pom.xml	13/04/2026 10:24 am	XML File	2 KB
target	13/04/2026 10:26 am	File folder	
README.md	13/04/2026 10:28 am	Markdown Source ...	1 KB



src/main/java/com/student/model/Student.java

```java
package com.student.model;

public class Student {
   private String name;
   private int catMarks;
   private int assignmentMarks;
   private int fatMarks;

   public Student(String name, int catMarks, int assignmentMarks, int fatMarks) {
      this.name = name;
      this.catMarks = catMarks;
      this.assignmentMarks = assignmentMarks;
      this.fatMarks = fatMarks;
   }

   public String getName() {
      return name;
   }

   public int getTotalMarks() {
      return catMarks + assignmentMarks + fatMarks;
   }

   public double getGPA() {
      int total = getTotalMarks();

      if (total >= 90) return 10.0;
      else if (total >= 80) return 9.0;
      else if (total >= 70) return 8.0;
      else if (total >= 60) return 7.0;
      else if (total >= 50) return 6.0;
      else return 5.0;
   }

   public String getGrade() {
      int total = getTotalMarks();

      if (total >= 90) return “A+”;
      else if (total >= 80) return “A”;
      else if (total >= 70) return “B”;
      else if (total >= 60) return “C”;
      else if (total >= 50) return “D”;
      else return “F”;
   }
}
```

Src/main/java/com/student/service/AnalyticsService.java

```java
package com.student.service;

import com.student.model.Student;
import java.util.List;

public class AnalyticsService {

    public Student findTopper(List<Student> students) {
        Student topper = students.get(0);

        for (Student s : students) {
            if (s.getTotalMarks() > topper.getTotalMarks()) {
                topper = s;
            }
        }
        return topper;
    }

    public double calculateAverageMarks(List<Student> students) {
        int sum = 0;

        for (Student s : students) {
            sum += s.getTotalMarks();
        }

        return (double) sum / students.size();
    }

    public void printReport(List<Student> students) {
        System.out.println("\n---- Student Report ----");

        for (Student s : students) {
            System.out.println(
                s.getName() + " | Total: " + s.getTotalMarks() +
                " | GPA: " + s.getGPA() +
                " | Grade: " + s.getGrade()
            );
        }
    }
}
```

src/main/java/com/student/Main.java

```java
package com.student;

import com.student.model.Student;
import com.student.service.AnalyticsService;

import java.util.ArrayList;
import java.util.List;

public class Main {

   public static void main(String[] args) {

      List<Student> students = new ArrayList<>();

      // Hardcoded data (for Jenkins compatibility)
      students.add(new Student("A", 25, 18, 45));
      students.add(new Student("B", 20, 15, 40));
      students.add(new Student("C", 28, 19, 48));
      students.add(new Student("D", 22, 17, 42));
      students.add(new Student("E", 26, 18, 46));

      AnalyticsService service = new AnalyticsService();

      service.printReport(students);

      Student topper = service.findTopper(students);
      System.out.println("\nTopper: " + topper.getName());

      double avg = service.calculateAverageMarks(students);
      System.out.println("Average Marks: " + avg);
   }
}
```

src/test/java/com/student/AnalyticsServiceTest.java

```java
package com.student;

import com.student.model.Student;
import com.student.service.AnalyticsService;
import org.junit.jupiter.api.Test;

import java.util.Arrays;
import java.util.List;

import static org.junit.jupiter.api.Assertions.*;
```

```
public class AnalyticsServiceTest {

  @Test
  public void testTopper() {
    List<Student> students = Arrays.asList(
        new Student("A", 20, 15, 40),
        new Student("B", 25, 18, 45),
        new Student("C", 10, 10, 20)
    );

    AnalyticsService service = new AnalyticsService();
    Student topper = service.findTopper(students);

    assertEquals("B", topper.getName());
  }

  @Test
  public void testAverage() {
    List<Student> students = Arrays.asList(
        new Student("A", 20, 15, 40),
        new Student("B", 25, 18, 45)
    );

    AnalyticsService service = new AnalyticsService();
    double avg = service.calculateAverageMarks(students);

    assertTrue(avg > 0);
  }
}
```

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
     xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
     xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
     http://maven.apache.org/xsd/maven-4.0.0.xsd">

  <modelVersion>4.0.0</modelVersion>

  <groupId>com.student</groupId>
  <artifactId>student-analytics</artifactId>
  <version>1.0-SNAPSHOT</version>

  <properties>
    <maven.compiler.source>17</maven.compiler.source>
    <maven.compiler.target>17</maven.compiler.target>
  </properties>
```

```
  <dependencies>
    <dependency>
      <groupId>org.junit.jupiter</groupId>
      <artifactId>junit-jupiter</artifactId>
      <version>5.9.3</version>
      <scope>test</scope>
    </dependency>
  </dependencies>

  <build>
    <plugins>
      <plugin>
        <groupId>org.apache.maven.plugins</groupId>
        <artifactId>maven-surefire-plugin</artifactId>
        <version>3.0.0-M7</version>
      </plugin>
    </plugins>
  </build>

</project>
```

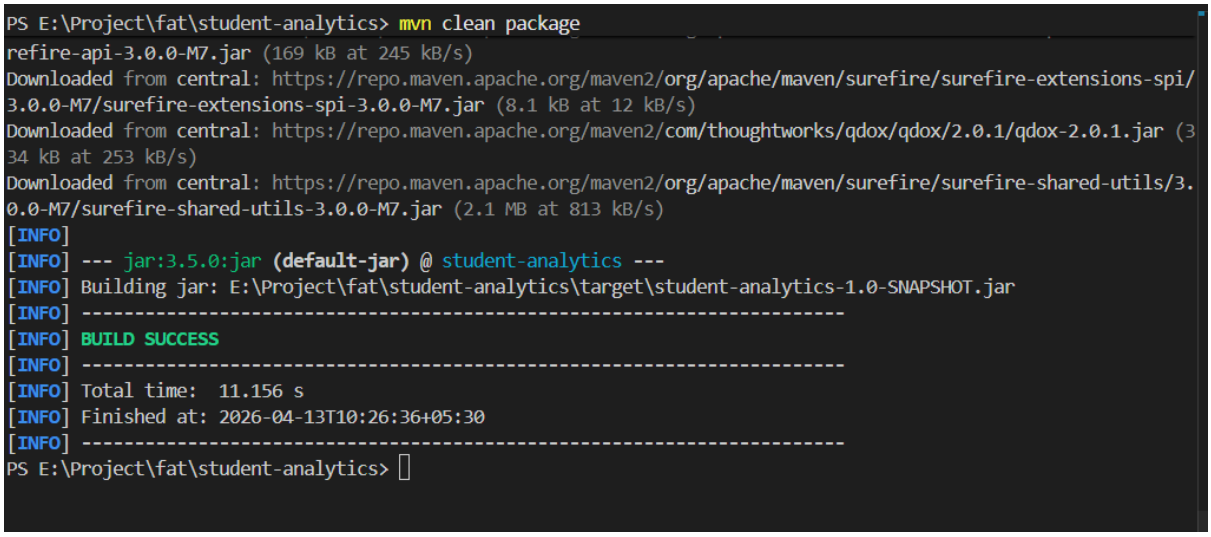
Jenkinsfile

```
pipeline {
  agent any

  tools {
    maven 'Maven'
  }

  stages {

    stage('Checkout') {
      steps {
        git branch: 'main', url: 'https://github.com/gunatejakp/student-analytics.git'
      }
    }

    stage('Build') {
      steps {
        bat 'mvn clean package'
      }
    }

    stage('Test') {
      steps {
```

```
            bat 'mvn test'
        }
      }
    }

    post {
      success {
        echo 'Build Successful 🎉'
      }
      failure {
        echo 'Build Failed ❌'
      }
    }
}
```

Screenshots:

Maven Build Successfully



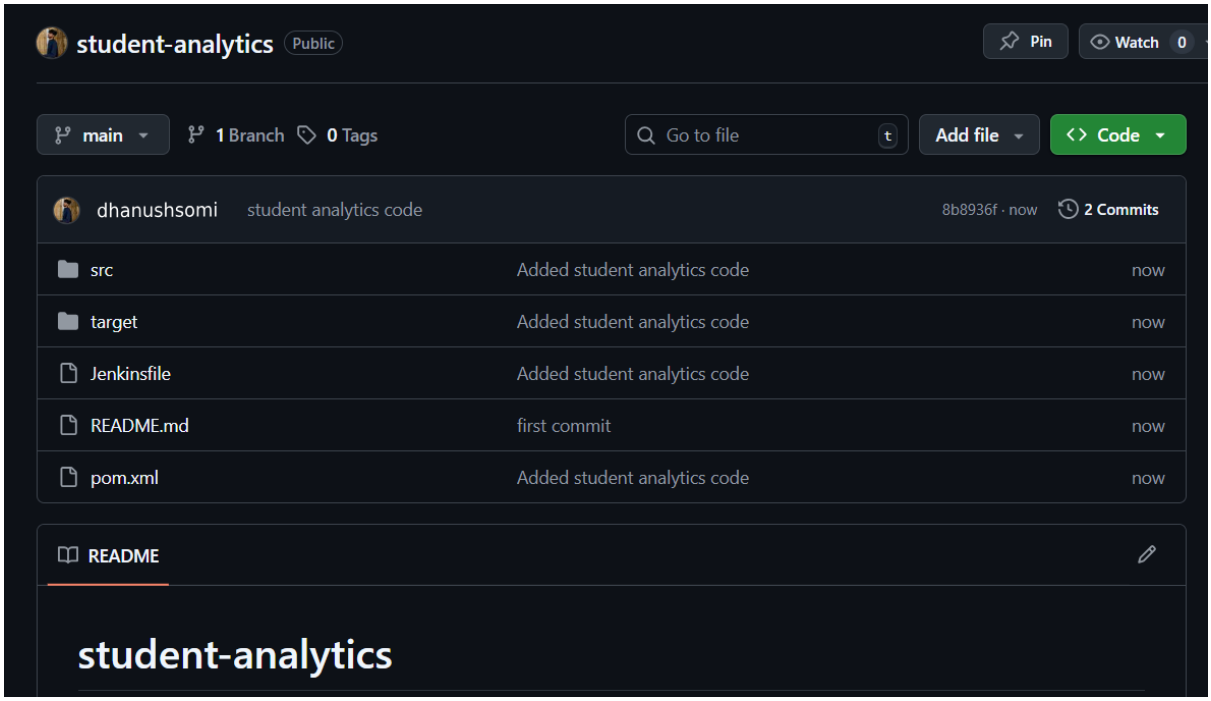
```
PS E:\Project\fat\student-analytics> mvn clean package
refire-api-3.0.0-M7.jar (169 kB at 245 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire-extensions-spi/
3.0.0-M7/surefire-extensions-spi-3.0.0-M7.jar (8.1 kB at 12 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/com/thoughtworks/qdox/qdox/2.0.1/qdox-2.0.1.jar (3
34 kB at 253 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire-shared-utils/3.
0.0-M7/surefire-shared-utils-3.0.0-M7.jar (2.1 MB at 813 kB/s)
[INFO]
[INFO] --- jar:3.5.0:jar (default-jar) @ student-analytics ---
[INFO] Building jar: E:\Project\fat\student-analytics\target\student-analytics-1.0-SNAPSHOT.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  11.156 s
[INFO] Finished at: 2026-04-13T10:26:36+05:30
[INFO] ------------------------------------------------------------------------
PS E:\Project\fat\student-analytics>
```

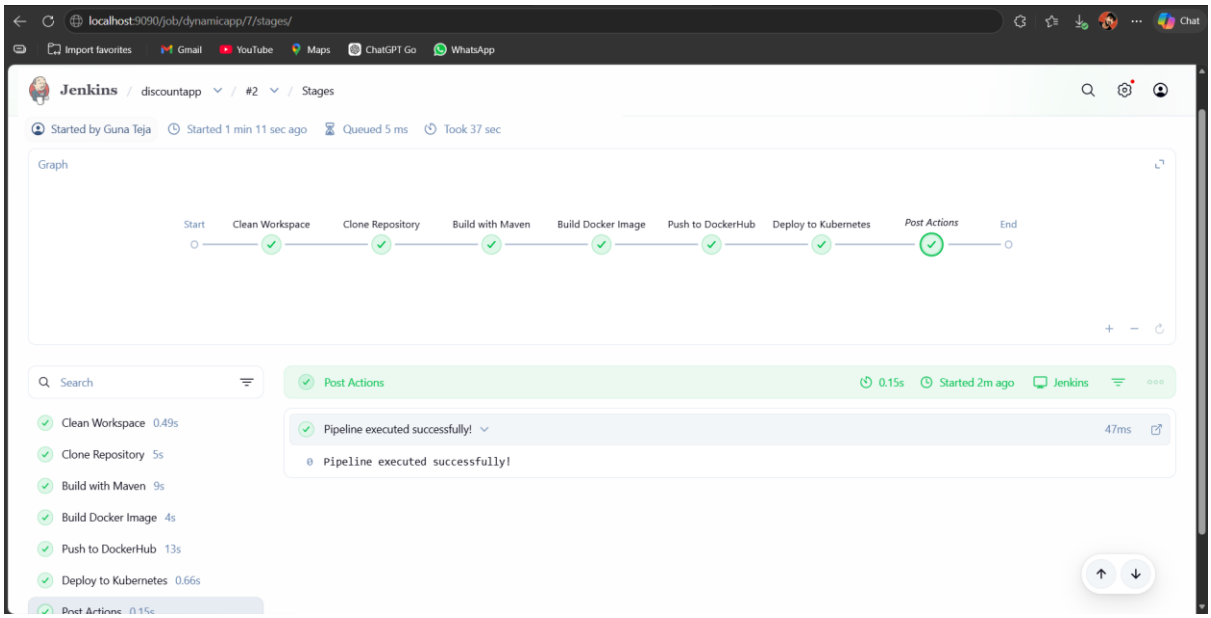
Cloning Repo in Github

```
PS E:\Project\fat\student-analytics> echo "# student-analytics" >> README.md
>> git init
>> git add README.md
>> git commit -m "first commit"
>> git branch -M main
>> git remote add origin https://github.com/gunatejakp/student-analytics.git
>> git push -u origin main
Initialized empty Git repository in E:/Project/fat/student-analytics/.git/
[master (root-commit) 58d912a] first commit
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 README.md
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 252 bytes | 252.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/gunatejakp/student-analytics.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS E:\Project\fat\student-analytics> git add .
>> git commit -m "Added student analytics code"
>> git push
```

After Cloning the GitHub Repo is:

Folder Structure:



Pipeline Build Finished

Pipeline Overview